

Module 5: Technical Feature Dictionary

Executive Overview

This document provides a comprehensive specification of all engineered behavioral features generated during Module 5 of the Customer Personality Segmentation project. These metrics capture core aspects of demographic profiles, household dynamics, purchasing frequency, channels, and deal responsiveness to enable robust clustering analysis.

Feature Name	Formula / Logic	Purpose & Business Significance	Expected ML Impact
Customer_Age	2026 - Year_Birth	Determines customer demographic generation and life-stage phase.	High variance predictor for product preference segmentation.
Customer_Tenure	Today - Dt_Customer	Measures duration of customer relationship and brand tenure in years.	Differentiates loyal legacy buyers from newly acquired users.
Total_Children	Kidhome + Teenhome	Calculates total dependent child count within the household.	Strongly correlates with category spending (e.g., family vs single).
Family_Size	Marital Status Map + Total Children	Determines overall household count (Single=1, Married=2 + kids).	Evaluates per-capita disposable income metrics.
Total_Spending	Sum(MntWines...MntGoldPr ods)	Monetary dimension (RFM framework). Total revenue generated.	Primary baseline feature for High-Value vs Low-Value clustering.
Total_Purchases	Sum(NumWeb, Catalog, Store)	Frequency dimension (RFM framework). Total transaction count.	Measures overall buying volume and activity density.
Avg_Spending_Per_Purchase	Total_Spending / Total_Purchases	Calculates average order value (AOV) per transaction.	Identifies premium buyers vs frequent micro-purchasers.
Digital_Engagement	NumWebPurchases + NumWebVisits	Aggregates online interaction and web traffic metrics.	Differentiates digital-first customers from retail-only buyers.
Deal_Dependency	NumDealsPurchases / Total_Purchases	Evaluates ratio of discounted transactions to total orders.	Isolates price-sensitive / bargain hunters from value buyers.
Total_Campaign_Acceptance	Sum(AcceptedCmp1..5 + Response)	Measures historical marketing offer acceptance responsiveness.	Identifies promo-sensitive target segments for future campaigns.

Implementation Guidelines

- **DataType Consistency:** All outputs are cast to Float64 / Int64 post-encoding.
- **Missing Value Strategy:** Numerical features imputed using median estimates prior to transformation.
- **Pipeline Compatibility:** Fully integrated into `CustomerPipelineTransformer` object in `src/preprocessing\_pipeline.py`.